

1) check whether a number is prime or not

$N = 30 \implies 1 \text{ and } 30$

1 to N

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 30

$N = 10$

(1) 2, 5, (10)

$N = 45$

1, 3, 5, 9, 15, 45

$N = 43$

1, 43

$N = 10$

1st factor $\implies 1$

last factor $\implies 10$

sqrt approach

45 $\implies 2$ to 44

45 $\implies \sqrt{45} \implies 6 \implies 2$ to 6

70 $\implies 2$ to 69

70 $\implies 2$ to 8

100 $\implies 2$ to 10

100 / 2

GCD(36, 24)

36 $\implies 1, 2, 3, 4, 6, 9, 12, 18, 36$

24 $\implies 1, 2, 3, 4, 6, 8, 12, 24$

GCD(36, 24)

36 - 24 $\implies 12$

GCD(12, 24)

24 - 12 $\implies 12$

GCD(12, 12)

$N = 100$

find all the prime no between

2 to N

GCD $\implies$ euclidean algorithm

2 to N $\implies$ sieve of erathosthenis

$N = 20$

2, 3, ~~4~~, ~~5~~, ~~6~~, ~~7~~, ~~8~~, ~~9~~, ~~10~~, ~~11~~, ~~12~~, 13, ~~14~~, ~~15~~, ~~16~~,
17, ~~18~~, ~~19~~, ~~20~~

$N = 10 \rightarrow 1 \leftarrow$
2, 3, 4, 5, 6, 7, 8, 9, 10

boolean array

cancel $\implies$ true

not cancel $\implies$ false



$N = 10 \implies 9$ eleme $\implies 11$ box

$N = 50 \implies 49$ eleme $\implies 51$ box

1111

primeNoTracker $\implies$

